

ExoVLA-WM: A Safety-Shielded Vision-Language-Action and World-Model Framework for Personalized Lower-Limb Exoskeleton Control

Ayush Debnath

Department of Mechanical Engineering
Indian Institute of Technology Kharagpur
Kharagpur, India
ayush.d@kgpian.iitkgp.ac.in

Abstract—Lower-limb exoskeletons must infer user intent, understand terrain, coordinate multiple powered joints, and remain safe under uncertainty. Existing systems typically separate locomotion-mode recognition from low-level control and operate over a fixed set of terrains. In parallel, vision-language-action (VLA) models have demonstrated broad semantic reasoning and cross-task transfer in manipulation, but their direct use for a human-worn robot is unsafe and mismatched to the dynamics, sensing, and action spaces of gait assistance. This paper proposes ExoVLA-WM, a hierarchical architecture that adapts a pretrained VLA to multimodal exoskeleton observations while restricting it to high-level locomotion decisions and bounded residual assistance. A predictive latent world model evaluates candidate action chunks before execution, and a deterministic safety shield enforces joint, torque, contact, stability, latency, and uncertainty constraints. The framework combines waist-mounted RGB-D perception, proprioception, foot-contact sensing, language commands, a gait-state estimator, an exoskeleton-specific action head, model-predictive risk scoring, and a real-time impedance controller. We specify the hardware layout, data format, training pipeline, simulation-to-hardware curriculum, safety mechanisms, human-in-the-loop personalization procedure, and a staged experimental protocol. Because no physical experiments have yet been completed, this manuscript is intentionally presented as a pre-experimental systems paper: it states testable hypotheses, baselines, ablations, and result templates without fabricating empirical outcomes.

Index Terms—vision-language-action models, world models, lower-limb exoskeleton, wearable robotics, computer vision, safe learning, human-in-the-loop optimization, locomotion intent

I. INTRODUCTION

Powered lower-limb exoskeletons couple a robot directly to the human body. A controller must therefore solve several problems at once: infer what the wearer intends to do, perceive the walking environment before contact, generate coordinated bilateral motion, adapt assistance to the individual, and maintain safety when perception or learning fails. Classical exoskeleton control is often hierarchical, with locomotion recognition and finite-state logic above trajectory, impedance, or torque control [1]. Such structures are interpretable and compatible with real-time guarantees, but they are commonly limited to

predefined activities and require substantial tuning across users and terrains.

Computer vision can improve predictive terrain awareness. Environment classifiers have been developed for robotic prostheses and exoskeletons [2], while Exosense demonstrated visual-inertial mapping and terrain reconstruction on an exoskeleton platform [3]. However, terrain labels alone do not provide language grounding, open-vocabulary goal understanding, or a unified representation of semantic context, body state, and action.

VLA models such as RT-2 [4], OpenVLA [5], and π_0 [6] map visual and language inputs to robot actions. OpenVLA was pretrained on large-scale, cross-embodiment manipulation data [7], and optimized adaptation recipes have introduced continuous action heads and action chunking [8]. These models are attractive because an exoskeleton user or therapist could issue commands such as “walk to the chair,” “climb these stairs slowly,” or “give more support on the left.” Yet existing VLA action spaces are designed primarily for manipulators, and an unconstrained model must not command motor voltages or unrestricted joint torques on a wearable robot.

We propose ExoVLA-WM, illustrated in Fig. 1. The core design principle is strict separation of responsibilities:

- the VLA interprets scene, language, and body context and outputs a restricted locomotion command;
- a learned world model predicts future body, contact, and risk variables for candidate command chunks;
- a deterministic safety layer rejects or projects unsafe commands;
- a verified real-time controller executes safe references and maintains a fallback behavior independently of the VLA.

The proposed contributions are:

- 1) an exoskeleton-specific multimodal VLA observation and action interface;
- 2) a predictive world-model layer for counterfactual gait-risk evaluation;

- 3) a safety shield combining state-machine logic, uncertainty gating, hard limits, and control-barrier-function constraints;
- 4) a staged data-collection and training protocol that starts with passive sensing and expert control before any learned residual is enabled;
- 5) a personalization method that optimizes assistance while preserving safety envelopes; and
- 6) a complete experimental design with baselines, ablations, metrics, and non-fabricated result templates.

II. RELATED WORK

A. Lower-Limb Exoskeleton Control

Lower-limb exoskeleton controllers are commonly organized into high-, mid-, and low-level layers [1]. High-level modules estimate intent or select an activity; mid-level modules produce joint references or assistance profiles; low-level modules regulate position, velocity, torque, or impedance. Locomotion recognition has used inertial sensing, EMG, force sensing, and multimodal fusion. Subject-adaptive classifiers [9] and neural models using a shank-mounted IMU [10] show that continuous gait phase and terrain-related variables can be estimated with lightweight sensors. More recent on-sensor approaches further motivate a separate low-power fallback recognizer [11].

Data-driven exoskeleton control is also emerging. Vianello et al. infer locomotion features and map therapist edits to desired posture and stiffness [12]. Simulation-trained policies can reduce the cost and risk of hardware data collection [13]. These works motivate learning-rich controllers, but they do not directly address open-vocabulary language grounding or the use of large pretrained VLA representations.

B. Visual Environment Understanding

Vision can identify terrain before physical contact and therefore reduce delayed controller transitions. Deep convolutional models have been studied for environment classification [2]. Exosense extends perception from classification to visual-inertial mapping and terrain reconstruction [3]. Our framework uses these ideas as a perception substrate, but combines geometry with language, proprioception, predictive action evaluation, and explicit uncertainty handling.

C. Vision-Language-Action Models

RT-2 introduced action tokenization within a vision-language model [4]. Open X-Embodiment consolidated heterogeneous robot datasets [7]; OpenVLA used such data to create an open, adaptable VLA [5]. OpenVLA-OFT showed that continuous action representations, parallel decoding, and action chunking can substantially improve adaptation speed and success [8]. Flow-based policies such as π_0 provide another route to continuous action generation [6]. Diffusion Policy [14] and ACT [15] are strong from-scratch baselines for sequence-level visuomotor control.

The proposed exoskeleton application differs in three important respects. First, the robot is physically attached to a human, so safety requirements are more stringent. Second, task success is not sufficient; interaction force, comfort, stability,

assistance timing, and user adaptation are also outcomes. Third, locomotion control requires deterministic high-rate loops that cannot depend on a large VLA inference stack.

D. World Models, Safety and Failure Detection

World models predict future outcomes of candidate actions and support planning in latent space. DreamerV3 [16] and TD-MPC2 [17] demonstrate scalable model-based learning and latent trajectory optimization. We adapt the concept to short-horizon prediction of gait phase, contacts, pelvis motion, clearances, joint states, interaction forces, and risk.

VLA safety is an active research area. SafeVLA formulates constrained learning for VLA policies [18], while SAFE uses internal VLA features to detect likely failures [19]. For a wearable robot, learned safety methods should supplement rather than replace deterministic protections. Control barrier functions (CBFs) provide a principled method for projecting proposed controls into a safe set [20], [21]; discrete CBF variants have also been applied to bipedal navigation [22].

III. SYSTEM REQUIREMENTS AND HARDWARE INTEGRATION

A. Mechanical Platform Assumptions

We assume a bilateral lower-limb device with powered hip and knee flexion-extension joints, rigid thigh and shank links, adjustable cuffs, foot interfaces, and a waist structure. The presented architecture remains applicable to passive ankle joints or additional powered degrees of freedom, but each new actuator increases the action dimension and safety-analysis burden.

The CAD rendering in Fig. 2 is used only to communicate integration locations. Final dimensions, transmission ratios, motor ratings, battery capacity, joint limits, and material choices must be derived from mechanical analysis and validated independently. Learning must not compensate for an under-designed frame, insufficient brakes, excessive backlash, poor alignment, or inadequate emergency support.

B. Sensor and Compute Stack

Table I specifies functional requirements rather than mandating particular vendors. A forward RGB-D or stereo camera should be mounted on the rigid pelvis frame and tilted downward to view the near walking surface. A second downward-facing camera improves stair-edge and foot-placement geometry. The pelvis IMU estimates global body orientation and acceleration. Absolute joint encoders, motor-side encoders, current sensing, and optional output torque sensors provide proprioception. Foot pressure sensors or load cells estimate heel strike, foot flat, toe-off, and left-right load transfer. Optional cuff force sensors and thigh or shank IMUs improve interaction and intent estimation.

The architecture uses two computing domains. The high-level computer may run a Linux-based perception and learning stack, but is not trusted for deterministic timing. The low-level controller uses a real-time operating environment or microcontroller and remains able to enter a safe state if the

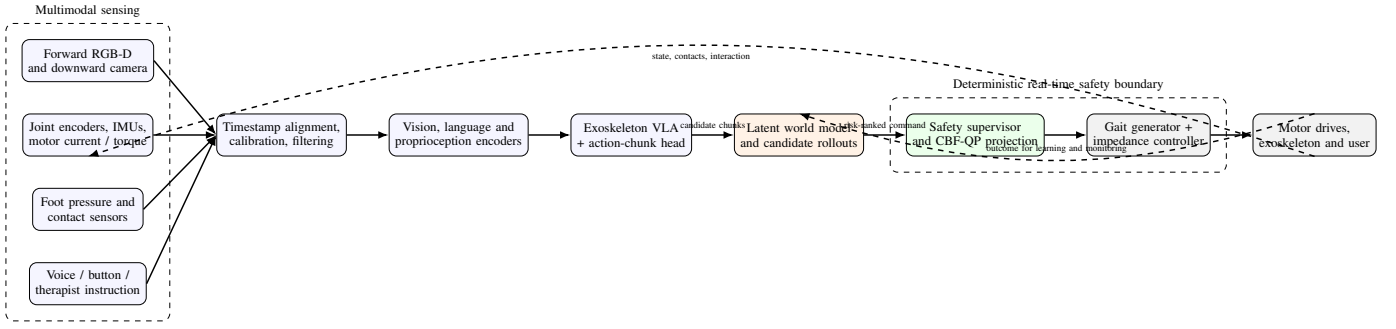


Fig. 1. Proposed ExoVLA-WM architecture. The VLA does not directly drive motors. It proposes restricted high-level action chunks; a world model predicts outcomes; a deterministic shield validates or modifies the command; and a real-time controller executes the result.

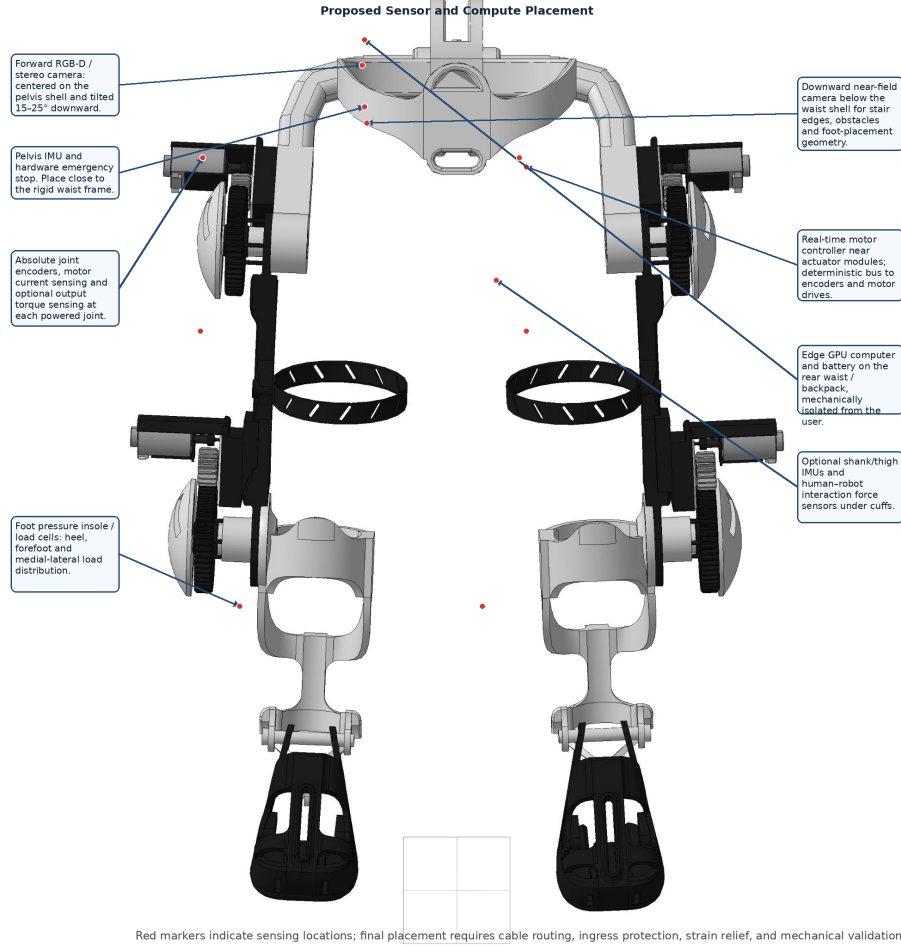


Fig. 2. Proposed sensor and compute placement on the supplied exoskeleton concept. Red markers denote sensing locations. The edge computer and battery are placed on the rear waist or backpack, while deterministic motor controllers remain close to the actuators.

high-level computer becomes unavailable. A watchdog must detect stale commands, invalid checksums, timing overruns, sensor disagreement, and communication loss.

C. Calibration and Time Synchronization

Before data collection, all cameras are calibrated intrinsically and extrinsically relative to a pelvis frame $\mathcal{F}_P$. Joint zero positions are measured using mechanical fixtures, and IMU

orientation is aligned to the link frame. Pressure sensors are zeroed and calibrated under known loads. Every sensor packet receives a monotonic timestamp. The dataset loader resamples observations to a common high-level clock and stores the original timestamps so that latency can be reconstructed.

Recommended calibration checks are:

- 1) camera reprojection error and depth-plane error;
- 2) joint-angle agreement between absolute and motor-side

TABLE I
FUNCTIONAL HARDWARE REQUIREMENTS AND TARGET UPDATE RATES

Module	Placement	Target rate	Primary purpose
Forward RGB-D / stereo camera	Center of rigid waist shell, downward tilt	15–30 Hz	Terrain class, depth, stair and obstacle geometry, open-vocabulary scene semantics
Downward camera	Under waist shell	30–60 Hz	Near-field edge detection, toe clearance region, landing region
Pelvis IMU	Near waist-frame center	200–1000 Hz	Pelvis roll, pitch, yaw rate, acceleration and instability detection
Joint encoders	Every powered joint	500–2000 Hz	Joint position and velocity; independent hardware limits
Current / torque sensing	Motor or joint output	500–2000 Hz	Interaction estimate, overload detection, torque control
Foot pressure / load cells	Heel, forefoot and optional medial-lateral zones	100–500 Hz	Contact state, gait phase, load transfer and slip cues
Optional cuff force sensors	Thigh and shank interfaces	100–1000 Hz	Human-robot interaction force and discomfort proxy
Edge GPU computer	Rear waist / backpack	VLA: 5–20 Hz; world model: 20–100 Hz	Vision, language, multimodal policy, predictive rollouts
Real-time controller	Near motor drives	500–2000 Hz	Safety supervision, gait generation, impedance / torque loops
Hardware emergency stop	Reachable by wearer and assistant	Asynchronous	Independent motor-disable path and brake engagement

encoders;

- 3) static IMU gravity-vector alignment;
- 4) left-right pressure symmetry under known standing loads;
- 5) end-to-end timing from exposure to action availability;
- 6) emergency-stop latency measured independently of software.

IV. PROBLEM FORMULATION

At high-level time step t , the observation is

$$\mathcal{O}_t = \{I_{t-k:t}^f, I_{t-k:t}^d, D_{t-k:t}, s_{t-k:t}, \ell_t, p_u\}, \quad (1)$$

where I^f and I^d are forward and downward image histories, D is depth, s contains proprioception and contact state, ℓ_t is a language instruction, and p_u is a user profile containing only approved non-sensitive control parameters such as leg lengths, mass range, preferred speed, assistance limits, and fitted joint range.

The VLA does not output raw motor voltage. It outputs a high-level action chunk

$$A_t^H = \{m_{t:t+H}, v_{t:t+H}^*, L_{t:t+H}^*, h_{t:t+H}^*, g_{t:t+H}^L, g_{t:t+H}^R, r_{t:t+H}\}, \quad (2)$$

where m is a discrete locomotion mode, v^* is desired speed, L^* is step length, h^* is step height or clearance, g^L, g^R are left-right assistance gains, and r is an optional bounded residual. The allowed mode set is

$$\mathcal{M} = \{\text{STOP, STAND, LEVEL, RAMP_UP, RAMP_DOWN, STAIR_UP, STAIR_DOWN, TURN_L, TURN_R, SIT}\}. \quad (3)$$

The low-level controller maps an accepted command to joint references q^{ref} and nominal torques

$$\tau_{nom} = K_p(q^{ref} - q) + K_d(\dot{q}^{ref} - \dot{q}) + \tau_{ff}. \quad (4)$$

The learned residual is saturated componentwise:

$$\tau_{des} = \tau_{nom} + \alpha \text{clip}(r, -\bar{r}, \bar{r}), \quad 0 \leq \alpha \leq \alpha_{max}. \quad (5)$$

The initial system sets $\alpha = 0$; nonzero residual assistance is enabled only after all higher-level experiments pass predefined safety gates.

V. EXOVLA-WM ARCHITECTURE

A. Multimodal Encoders

A pretrained VLA provides a vision-language backbone. Image tokens are generated from the forward and downward views. A dedicated proprioception projector encodes joint positions, velocities, motor currents, pelvis IMU, pressure features, gait phase, and previous mode. The projected proprioceptive tokens are inserted using learned modality embeddings and temporal positions. A user-profile token provides permitted personalization parameters without exposing raw medical records.

We recommend three initial model variants:

- 1) **OpenVLA adapter**: freeze most of the VLA and train LoRA layers, a proprioception projector, and an exoskeleton action head;
- 2) **OFT-style continuous head**: predict continuous chunks with parallel decoding and an L_1 objective following the design lessons of OpenVLA-OFT [8];
- 3) **from-scratch baseline**: train ACT or Diffusion Policy on the same observations to determine whether VLA pretraining adds value.

B. Hierarchical Action Head

The action head has three branches:

- categorical logits for locomotion mode;
- continuous chunk regression for speed, step geometry, and assistance gains;

- calibrated uncertainty and failure likelihood.

The supervised imitation loss is

$$\mathcal{L}_{VLA} = \lambda_m \mathcal{L}_{CE}(m, \hat{m}) + \lambda_c \|a_c - \hat{a}_c\|_1 + \lambda_s \mathcal{L}_{smooth} + \lambda_u \mathcal{L}_{cal}, \quad (6)$$

where a_c denotes continuous chunk parameters. The smoothness term penalizes abrupt changes in speed, assistance, and residual commands. The calibration term may combine temperature scaling, Brier loss, and failure-label supervision. Failure detection can additionally use internal VLA features, following the motivation of SAFE [19].

C. Latent World Model

The world model encodes the current multimodal state into z_t and predicts future latent state, contact, kinematics, interaction, and risk under candidate action chunks:

$$z_{t+1} \sim f_\theta(z_t, A_t^H), \quad (7)$$

$$\hat{y}_{t+1} = g_\theta(z_{t+1}), \quad (8)$$

where $\hat{y}$ contains predicted pelvis orientation, joint state, foot contacts, toe clearance, interaction torque, collision indicator, slip indicator, and confidence. Training uses a weighted objective

$$\mathcal{L}_{WM} = \lambda_z \mathcal{L}_{dyn} + \lambda_q \mathcal{L}_{kin} + \lambda_c \mathcal{L}_{contact} + \lambda_f \mathcal{L}_{force} + \lambda_r \mathcal{L}_{risk}. \quad (9)$$

At inference, the system samples or enumerates a small set of candidate chunks near the VLA proposal. Each candidate receives a horizon cost

$$J(A) = \sum_{i=1}^H \left[w_e c_{task} + w_s c_{stability} + w_j c_{joint} + w_f c_{force} + w_o c_{obstacle} + w_u c_{uncertainty} + w_E c_{energy} \right]_{t+i}. \quad (10)$$

The lowest-risk feasible candidate is passed to the deterministic safety layer. If all candidates are infeasible, the system selects a safe stop or support state.

D. Uncertainty Estimation

A single confidence score is insufficient. We propose reporting:

- predictive entropy over locomotion mode;
- ensemble variance or dropout variance for continuous action parameters;
- disagreement between the VLA terrain interpretation and a lightweight independent classifier;
- world-model rollout variance;
- out-of-distribution score in visual-proprioceptive feature space;
- calibrated failure probability.

Uncertainty metrics for VLA-enabled robots are still developing [23]. Therefore, thresholds must be tuned using held-out failure scenarios, not assumed to transfer from manipulation benchmarks.

VI. DETERMINISTIC SAFETY SUPERVISOR

A. State Machine and Gating

The safety supervisor maintains modes BOOT, CALIBRATE, STANDBY, ACTIVE, TRANSITION, SAFE_STOP, and FAULT. Learned commands are accepted only in ACTIVE or approved TRANSITION states. A transition guard checks contact pattern, gait phase, terrain confidence, user command consistency, and minimum available stopping margin.

Immediate fallback is triggered by any of the following:

- hardware emergency stop;
- stale high-level command or missed watchdog deadline;
- joint position, velocity, current, torque, or temperature beyond a warning boundary;
- inconsistent redundant encoders;
- excessive pelvis tilt or angular rate;
- unexpected loss of support contact;
- sensor dropout, invalid depth, or excessive VLA/world-model uncertainty;
- predicted collision, insufficient toe clearance, or unsafe load transfer;
- communication corruption or controller reset.

B. Control Barrier Function Projection

Let the robot-human state be x and the control input be joint torque τ . Define safety functions $h_i(x) \geq 0$ for joint limits, velocity limits, torque margins, pelvis orientation, toe clearance, and contact-dependent constraints. The desired torque is projected through

$$\tau^* = \arg \min_{\tau, \delta} \frac{1}{2} \|\tau - \tau_{des}\|_W^2 + \rho \|\delta\|^2 \quad (11)$$

$$\text{s.t. } \dot{h}_i(x, \tau) + \gamma_i h_i(x) \geq -\delta_i, \quad \forall i, \quad (12)$$

$$\tau_{min} \leq \tau \leq \tau_{max}, \quad |\Delta\tau| \leq \Delta\tau_{max}, \quad (13)$$

$$\delta_i \geq 0. \quad (14)$$

This formulation follows the CBF-QP principle [20]. Because the coupled human-exoskeleton dynamics are uncertain, learned model corrections may be added only with conservative margins and validation, consistent with learning-based CBF work [21]. If the optimization is infeasible or exceeds its deadline, the controller uses a pre-certified fallback torque.

C. Safety Limits

Numerical limits must be derived from the actual mechanism, user population, actuator capability, clinical input, and formal risk analysis. The source package therefore exposes configuration variables but intentionally does not assign universal values. Each limit has a warning threshold, a hard threshold, a debounce duration, and a documented fallback. Safety must not rely on a single software process.

VII. DATA COLLECTION AND DATASET DESIGN

A. Staged Data Collection

Data collection proceeds in six gates.

Gate 0: Bench and fixture tests. Validate motors, brakes, encoders, current limits, watchdogs, bus timing, and emergency stop without a human.

Gate 1: Passive sensing. A participant wears the unpowered or transparent-control device while walking with external support. Record sensor data and labels without allowing learned control.

Gate 2: Expert controller. Use an established finite-state gait controller or therapist-operated reference system. The learned model observes but does not act.

Gate 3: Shadow mode. Run VLA and world-model inference online; log proposed actions, uncertainty, and safety rejections while the expert controller remains in command.

Gate 4: High-level shared autonomy. Permit the learned system to select among validated gait primitives while the safety layer and expert low-level controller remain authoritative.

Gate 5: Bounded residual. Enable a small residual assistance term only after all prior gates meet predefined safety and reliability criteria.

B. Episode Schema

Each episode stores:

```
episode_id, participant_code
session_id, calibration_version
hardware_version, software_commit
rgb_forward, rgb_down, depth
camera_timestamps
joint_position, joint_velocity
motor_current, torque
pelvis_imu, optional_limb_imus
foot_pressure
expert_mode, gait_phase, terrain
transition_label
language_instruction
therapist_override
high_level_action
low_level_reference, executed_torque
safety_flags, uncertainty
failure_annotation
outcome_metrics, user_feedback
```

Personally identifying information should be stored separately under approved governance. Video release requires explicit consent and de-identification procedures.

C. Language and Task Annotation

Language commands are grouped into intent families:

- locomotion: start, stop, slow, speed up, turn;
- terrain: climb stairs, descend stairs, use ramp, step over obstacle;
- semantic goal: go to the chair, stop at the doorway;
- assistance: more or less support, left-right adjustment;
- safety: stop now, hold position, ask for help.

Paraphrases should be recorded to test language generalization, but every open-vocabulary command must map to a restricted machine-readable command before execution.

VIII. TRAINING PIPELINE

A. Stage A: Perception and Gait-State Pretraining

Train separate modules for terrain segmentation, depth-derived geometry, contact state, and gait phase. These modules provide interpretable diagnostics and independent checks. Terrain labels include level ground, ramp up/down, stair up/down, obstacle, drop-off, and unknown. Geometry targets include distance to first stair edge, riser height, tread depth, ramp slope, obstacle height, and candidate landing regions.

B. Stage B: VLA Adaptation by Imitation

Initialize from OpenVLA or another available VLA. Train a proprioception projector and high-level action head with parameter-efficient tuning. Balance the dataset by participant, activity, transition, terrain geometry, lighting, and instruction paraphrase. Use sequence windows rather than individual frames. Strong augmentations are applied only when their physical meaning is preserved; depth corruption and camera occlusion are included as explicit failure cases.

C. Stage C: World-Model Training

Train the world model on expert and shadow-mode trajectories. To reduce unsafe extrapolation, initial rollouts are short and candidates stay close to expert actions. Use ensembles or bootstrap heads to estimate epistemic uncertainty. Oversample transitions, near-misses, emergency stops, and sensor failures. A candidate is labeled unsafe when it violates a hard limit, triggers the supervisor, produces insufficient clearance, causes unexpected contact, or is rejected by an expert.

D. Stage D: Simulation and Offline Improvement

A physics simulation should represent the exoskeleton, a human or musculoskeletal surrogate, actuator dynamics, transmission compliance, sensor noise, terrain variation, and communication delay. Domain randomization covers body dimensions, mass distribution, friction, motor parameters, camera pose, latency, and sensor bias. Simulation may train residual policies or expand the world-model dataset, but deployment remains gated by real-data uncertainty. Experiment-free simulation learning has shown promise for exoskeleton assistance [13]; nevertheless, human-worn validation must proceed conservatively.

E. Stage E: Human-in-the-Loop Personalization

Personalization optimizes a low-dimensional assistance vector ϕ_u , not the entire VLA. Candidate parameters may include phase, amplitude, stiffness, damping, and left-right assistance ratio. The objective is

$$\phi_u^* = \arg \min_{\phi \in \Phi_{safe}} w_M \hat{M}(\phi) + w_I \hat{I}(\phi) + w_C \hat{C}(\phi) + w_A \|\phi - \phi_0\|^2, \quad (15)$$

where $\hat{M}$ is a metabolic or muscle-effort proxy, $\hat{I}$ is interaction force, $\hat{C}$ is discomfort or instability cost, and ϕ_0 is a conservative baseline. Human-in-the-loop optimization has shown that individual customization can matter [24], [25], but recent work also highlights variability and co-adaptation limitations [26].

Algorithm 1 Online ExoVLA-WM Control Loop

```

1: Initialize verified gait controller and safe-state machine
2: while system enabled do
3:   Read synchronized sensing packet  $\mathcal{O}_t$ 
4:   if hardware fault, stale data, or emergency stop then
5:     enter SAFE_STOP; continue
6:   end if
7:   Estimate gait phase, contact, terrain and independent
   confidence
8:   VLA proposes high-level action chunk  $\hat{A}_t^H$  and uncer-
   tainty
9:   Generate candidate set  $\mathcal{C}(\hat{A}_t^H)$  including safe stop
10:  for all  $A \in \mathcal{C}$  do
11:    Roll out world model; compute  $J(A)$  and feasibility
12:  end for
13:  Select lowest-cost feasible candidate  $A^*$ 
14:  if uncertainty or failure probability exceeds threshold
  then
15:    replace  $A^*$  with conservative fallback
16:  end if
17:  Generate nominal joint references and  $\tau_{des}$ 
18:  Solve safety projection in Eq. (14)
19:  if solver infeasible or late then
20:    execute certified fallback torque
21:  else
22:    execute  $\tau^*$  and log all decisions
23:  end if
24: end while

```

Therefore, the optimizer uses conservative trust regions and repeated validation rather than one-session convergence claims.

IX. IMPLEMENTATION PLAN

A. Software Processes

The implementation is divided into independently restartable processes:

- 1) sensor drivers and hardware timestamping;
- 2) camera preprocessing and depth geometry;
- 3) proprioceptive state estimation;
- 4) VLA inference server;
- 5) world-model rollout server;
- 6) safety supervisor and real-time controller;
- 7) logger, dashboard, and replay tools.

The safety controller must not block on a network call, GPU inference, file write, or visualization process. High-level messages contain sequence number, timestamp, validity interval, model version, uncertainty, and checksum.

B. Target Latency Budget

A plausible initial latency budget is shown in Table II. Values are engineering targets, not measured results.

The VLA uses receding-horizon chunks so that the low-level controller can continue tracking a valid reference between high-level updates. If the validity interval expires, the system

TABLE II
ILLUSTRATIVE HIGH-LEVEL LATENCY BUDGET

Component	Target
Camera exposure, transfer and preprocessing	< 40 ms
Depth / terrain geometry	< 25 ms
Proprioception and gait-state update	< 5 ms
VLA chunk generation	< 100 ms
World-model candidate scoring	< 20 ms
High-level safety decision	< 5 ms
Low-level controller cycle	0.5–2 ms

decelerates or enters a support state rather than replaying stale actions.

C. Model Sizes and Training Configuration

A practical first implementation freezes the majority of a 7B-scale VLA and trains low-rank adapters, the proprioception projector, and a compact action head. Mixed precision and quantization may reduce inference cost, but every quantized model must be revalidated. The world model can begin with a smaller recurrent state-space or transformer model because its horizon and output variables are restricted. Hyperparameters in the source code should be treated as starting points and tuned on a validation set grouped by participant.

X. EXPERIMENTAL EVALUATION

A. Research Questions

The study tests the following hypotheses:

H1:

Multimodal VLA adaptation improves generalization to unseen visual scenes and paraphrased instructions compared with a closed-set terrain classifier.

H2:

The world model reduces unsafe transition proposals and near-miss events compared with direct VLA-to-gait-primitive selection.

H3:

The deterministic safety shield reduces hard-limit and contact-consistency violations without causing unacceptable task failure or latency.

H4:

Personalized assistance improves at least one user-centered outcome without increasing interaction force or instability.

H5:

The proposed architecture degrades safely under camera occlusion, depth corruption, language ambiguity, communication delay, and sensor dropout.

B. Baselines

C. Staged Test Protocol

Phase 1: Simulation. Evaluate thousands of randomized episodes, including sensor faults and rare terrain transitions. No human exposure.

TABLE III
PLANNED BASELINES

Method	Vision/language	World model	Safety
B0 FSM	Closed-set sensors only	No	Hard limits
B1 CV+FSM	Terrain classifier / geometry	No	Hard limits
B2 ACT / Diffusion	Vision + state; optional language	No	Hard limits
B3 VLA-direct	VLA high-level command	No	Hard limits
B4 VLA+WM	VLA + predictive ranking	Yes	Hard limits
Ours	VLA + predictive ranking	Yes	Full supervisor + CBF-QP

Phase 2: Hardware-in-the-loop and dummy-load tests.

Use link fixtures, instrumented test rigs, and suspended leg surrogates. Verify timing, limit enforcement, and safe-stop behavior.

Phase 3: Able-bodied participants. Conduct supervised trials with overhead fall protection or parallel bars, an assistant, hardware emergency stop, conservative speed, and zero learned residual initially.

Phase 4: Target clinical population. Proceed only after engineering review, institutional ethics approval, clinician involvement, inclusion/exclusion criteria, and successful prior gates. The protocol must be adapted to the intended medical claim and local regulation.

Tasks include standing, starting, stopping, level walking, turning, ramp ascent/descent, stair ascent/descent, sit-to-stand, obstacle crossing, semantic goal following, and ambiguous-command rejection. Environmental variation includes lighting, surface texture, clutter, obstacle dimensions, camera occlusion, and unseen stair geometry within approved bounds.

D. Metrics

E. Ablations

Ablations remove one component at a time: language, depth, downward camera, foot pressure, pelvis IMU, proprioceptive history, action chunking, world model, failure detector, CBF projection, and personalization. Additional tests compare a unified VLA encoder against separate vision and intent models. The central question is not merely whether the full model succeeds, but which components add safety and generalization beyond a strong deterministic baseline.

F. Statistical Analysis

Use participant-wise train/validation/test splits and report confidence intervals. Human trials should use repeated-measures analysis because every participant experiences multiple controllers. A mixed-effects model can include controller as a fixed effect and participant as a random effect. Binary success outcomes can use mixed-effects logistic regression. Continuous metrics can use linear mixed models or appropriate nonparametric alternatives after residual checks. Multiple comparisons should be corrected. Effect sizes and confidence intervals are reported alongside p -values.

G. Result Tables Without Fabrication

Table V is a reporting template to complete after experiments.

XI. FAILURE SCENARIOS AND RISK CONTROLS

Formal risk management should follow the intended product category. ISO 13482 addresses personal-care robot hazards but explicitly does not cover robots classified as medical devices; medical use requires an appropriate medical-device risk process such as ISO 14971 [27], [28]. In the United States, powered lower-extremity exoskeletons are regulated as Class II devices under product code PHL and 21 CFR 890.3480 [29]. This paper is not a regulatory submission and does not establish compliance.

XII. ETHICS, PRIVACY AND REPRODUCIBILITY

Human studies require institutional ethics approval, informed consent, stopping rules, trained supervision, and a plan for adverse events. Participants must be told which portions of control are learned, what data are recorded, and when an operator may override the system. The system should minimize collection of identifiable video and separate identity from episode data. Voice commands may be transcribed locally and raw audio deleted when not required.

Reproducibility artifacts should include:

- mechanical and electrical interface specifications that do not expose unsafe build details;
- calibration procedures and sensor schemas;
- dataset splits by participant and environment;
- model configuration, training seeds, software commits, and evaluation scripts;
- safety configuration version and complete intervention logs;
- negative results and failure cases, not only successful demonstrations.

XIII. DISCUSSION

A. Why a Hierarchy Is Necessary

An end-to-end VLA-to-torque controller is conceptually simple but inappropriate as the first implementation. Large-model inference has variable latency, opaque failure modes, and training-data mismatch. A hierarchy confines semantic reasoning to a slow, restricted action space and preserves deterministic control at the physical interface. This also allows each layer to be evaluated independently.

B. Where Novelty Must Be Demonstrated

The work is not novel merely because a camera or a VLA is attached to an exoskeleton. Empirical novelty should be demonstrated through at least three axes:

- 1) an exoskeleton-specific VLA interface that fuses vision, language, proprioception, and contact history;
- 2) predictive world-model rejection that measurably reduces risky transitions or failure severity;
- 3) safe personalization that improves a human-centered metric without degrading stability or interaction safety.

TABLE IV
EVALUATION METRICS

Category	Metrics	Interpretation
Task performance	Completion rate, time, path deviation, transition success, instruction-following accuracy	Whether the requested mobility task is completed correctly
Perception	Terrain F1, geometry MAE, transition anticipation time, unknown detection AUROC	Whether the system understands terrain before contact
Safety	Hard-limit violations, supervisor interventions, near-falls, unexpected contacts, minimum toe clearance, stop distance	Whether failures are detected and contained
Control	Joint tracking RMSE, torque RMSE, torque-rate peaks, interaction force, gait-phase error	Quality and smoothness of physical execution
Human state	Pelvis roll/pitch, step symmetry, double-support time, optional EMG or metabolic cost	Stability, effort and biomechanical effect
Uncertainty	ECE, Brier score, failure AUROC, time-to-detection, selective success vs. coverage	Calibration and usefulness of abstention
Systems	End-to-end latency, deadline misses, power consumption, thermal throttling, dropped frames	Real-time feasibility and robustness
User outcomes	Comfort score, perceived safety, therapist workload, usability questionnaire	Practical acceptability
Generalization	Unseen users, terrains, lighting and paraphrases	Whether performance transfers beyond training conditions

TABLE V
PRIMARY RESULT TEMPLATE. NO NUMERICAL VALUES ARE CLAIMED IN THIS PRE-EXPERIMENTAL DRAFT.

Method	Task success	Transition F1	Safety interventions	Near-misses	Latency	Comfort
B0 FSM	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>
B1 CV+FSM	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>
B2 ACT / Diffusion	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>
B3 VLA-direct	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>
B4 VLA+WM	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>
ExoVLA-WM	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>	<i>To be measured</i>

A strong paper should also release a benchmark containing paired scene, instruction, body-state, action, risk, and outcome labels.

C. Limitations

The proposed architecture has important limitations. Manipulation-pretrained VLA representations may transfer poorly to gait. Real exoskeleton data are expensive and safety constrained. User behavior is nonstationary, and personalization may reflect adaptation rather than a stable optimum. World models can be confidently wrong outside the training distribution. CBF guarantees depend on model assumptions and reliable state estimation. Vision may fail under sunlight, blur, reflective surfaces, or crowd occlusion. Finally, clinical benefit cannot be inferred from able-bodied laboratory experiments.

XIV. CONCLUSION

This paper specified ExoVLA-WM, a complete research plan for integrating VLA models and predictive world models into

a bilateral lower-limb exoskeleton without giving the learned model unrestricted motor authority. The design combines multimodal sensing, language-grounded high-level action chunks, predictive risk evaluation, calibrated abstention, deterministic safety projection, conventional real-time control, and staged human-in-the-loop personalization. The proposed experiments are structured to determine whether VLA semantics and world-model foresight improve generalization and safety beyond strong finite-state, computer-vision, and imitation-learning baselines. The next required step is not additional architectural complexity, but disciplined implementation of bench validation, synchronized data collection, shadow-mode evaluation, and safety-gated human trials.

APPENDIX A MINIMUM BUILD AND INTEGRATION CHECKLIST

- 1) Verify structural factors of safety, mechanical stops, joint alignment, cuff comfort, pinch-point protection, transmission backlash, and brake behavior.

TABLE VI
PRELIMINARY FAILURE-MODE AND EFFECTS ANALYSIS

Failure mode	Possible effect	Detection	Control / fallback
Camera occlusion or glare	Wrong terrain or missed obstacle	Image-quality monitor, depth invalid ratio, model disagreement	Reduce speed, independent proprioceptive mode, safe stop before transition
Ambiguous language	Incorrect task selection	Low instruction confidence, contradiction with current state	Ask for confirmation or reject; never infer an irreversible action
VLA hallucination / OOD scene	Unsafe high-level command	OOD score, failure detector, world-model risk, rule inconsistency	Candidate rejection and conservative gait primitive
World-model error	False-safe rollout	Ensemble disagreement, validation residual monitor	Increase margins; safety supervisor remains authoritative
Pressure sensor failure	Incorrect gait phase / contact	Redundancy, plausibility checks, no-load drift	Use encoder/IMU fallback; block terrain transition
Encoder discrepancy	Wrong joint state	Absolute vs. motor encoder disagreement	Disable affected joint, brake or safe support state
High-level computer crash	No new command	Watchdog timeout	Continue bounded valid chunk briefly, then decelerate / safe stop
QP infeasibility or overrun	No certified projected torque	Solver status and timing monitor	Immediate verified fallback torque; log event
Unexpected user behavior	Instability or discomfort	Interaction force, pelvis IMU, contact mismatch	Reduce assistance, increase compliance, stop and request help
Battery or thermal issue	Loss of actuation or degraded torque	Voltage, current and temperature monitors	Controlled stop before critical threshold; mechanical support

- 2) Establish independent hardware current limits and emergency motor disable.
- 3) Install absolute encoders and confirm startup state without motion.
- 4) Route cables with strain relief and joint-motion clearance.
- 5) Mount cameras and IMUs to rigid members, not flexible covers.
- 6) Implement a deterministic communication protocol with timestamps, sequence numbers and watchdogs.
- 7) Validate safe-stop behavior under compute loss, sensor loss and battery reduction.
- 8) Collect passive and expert-control data before enabling learned decisions.
- 9) Run shadow mode and quantify false transitions, uncertainty calibration and safety rejections.
- 10) Enable high-level primitive selection before any residual torque.

APPENDIX B SUGGESTED TRAINING CONFIGURATION

The following values are starting points, not validated optima:

- sequence context: 0.5–2.0 s of synchronized history;
- high-level action horizon: 0.25–1.0 s;
- LoRA rank: 16–64 on selected attention and feed-forward layers;
- proprioception token dimension: 256–1024;
- optimization: AdamW, gradient clipping, cosine decay;
- balanced sampling by user, terrain, transition, instruction and failure type;
- validation split: unseen participants and unseen physical locations;
- early stopping: composite of action error, calibration and safety-event recall;

- world-model ensemble: 3–5 members or bootstrap heads;
- deployment: quantization only after accuracy, calibration and timing revalidation.

APPENDIX C SAFETY-GATED MILESTONES

TABLE VII
GO/NO-GO MILESTONES

Gate	Required evidence before progression
G0	Independent emergency stop, current limit and watchdog pass repeated fault-injection tests
G1	Sensor calibration and synchronization remain within specified tolerance
G2	Expert controller completes approved tasks with no unresolved safety event
G3	Shadow VLA has acceptable transition precision, calibrated abstention and bounded latency
G4	World-model risk ranking improves failure recall on held-out faults
G5	Shared autonomy meets intervention and near-miss thresholds under harness
G6	Residual assistance remains within certified bounds and improves a predefined metric

APPENDIX D EXAMPLE MACHINE-READABLE HIGH-LEVEL COMMAND

```
{
  "sequence": 18231,
  "timestamp_ns": 921834772110,
  "valid_until_ns": 921834972110,
  "mode": "STAIR_UP",
  "speed_mps": 0.22,
  "step_length_m": 0.28,
  "step_height_m": 0.17,
  "assist_left": 0.65,
  "assist_right": 0.62,
  "residual_enabled": false,
  "vla_uncertainty": 0.08,
```

```

"failure_probability": 0.03,
"world_model_risk": 0.11,
"model_version": "exovla_wm_v0.3",
"checksum": "...
}

```

The real implementation should use a deterministic binary message definition and authenticated software updates. The JSON representation is shown only for readability.

REFERENCES

- [1] R. Baud, A. R. Manzoori, A. Ijspeert, and M. Bouri, "Review of control strategies for lower-limb exoskeletons to assist gait," *Journal of NeuroEngineering and Rehabilitation*, vol. 18, no. 1, p. 119, 2021.
- [2] B. Laschowski, W. McNally, A. Wong, and J. McPhee, "Environment classification for robotic leg prostheses and exoskeletons using deep convolutional neural networks," *Frontiers in Neurobotics*, vol. 15, 2022.
- [3] J. Wang, M. Mattamala, C. Kassab, G. Burger, F. Elneceve, L. Zhang, M. Petriaux, and M. Fallon, "Exosense: A vision-based scene understanding system for exoskeletons," 2025.
- [4] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski, T. Ding, D. Driess, A. Dubey, C. Finn *et al.*, "Rt-2: Vision-language-action models transfer web knowledge to robotic control," in *Conference on Robot Learning*, 2023.
- [5] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi *et al.*, "Openvla: An open-source vision-language-action model," in *Conference on Robot Learning*, 2025.
- [6] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter *et al.*, " π_0 : A vision-language-action flow model for general robot control," in *Robotics: Science and Systems*, 2025.
- [7] Open X-Embodiment Collaboration, "Open x-embodiment: Robotic learning datasets and rt-x models," 2023.
- [8] M. J. Kim, C. Finn, and P. Liang, "Fine-tuning vision-language-action models: Optimizing speed and success," 2025.
- [9] J. Wang, D. Wu, Y. Gao, X. Wang, X. Li, G. Xu *et al.*, "Integral real-time locomotion mode recognition based on ga-cnn for lower limb exoskeleton," *Journal of Bionic Engineering*, 2022.
- [10] F. Weigand, A. Hohl, J. Zeiss, U. Konigorski, and M. Grimmer, "Continuous locomotion mode recognition and gait phase estimation based on a shank-mounted imu with artificial neural networks," 2022.
- [11] M. Razmi and I. Shojaei, "Event-driven on-sensor locomotion mode recognition using a shank-mounted imu with embedded machine learning for exoskeleton control," 2026.
- [12] L. Vianello, C. Lhoste, E. B. Kucuktabak, M. Short, L. Hargrove, and J. L. Pons, "Deep-learning control of lower-limb exoskeletons via simplified therapist input," 2024.
- [13] S. Luo, M. Jiang, S. Zhang, J. Zhu, S. Yu, I. D. Silva, T. Wang, E. Rouse, B. Zhou, H. Yuk *et al.*, "Experiment-free exoskeleton assistance via learning in simulation," *Nature*, vol. 630, no. 8016, pp. 353–359, 2024.
- [14] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," *The International Journal of Robotics Research*, 2024, arXiv:2303.04137.
- [15] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," 2023.
- [16] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering diverse domains through world models," 2023.
- [17] N. Hansen, H. Su, and X. Wang, "Td-mpc2: Scalable, robust world models for continuous control," in *International Conference on Learning Representations*, 2024.
- [18] B. Zhang, Y. Zhang, J. Ji, Y. Lei, Y. Cai, J. Dai, Y. Chen, and Y. Yang, "Safevla: Towards safety alignment of vision-language-action model via constrained learning," in *Advances in Neural Information Processing Systems*, 2025.
- [19] Q. Gu, Y. Ju, S. Sun, I. Gilitschenski, H. Nishimura, M. Itkina, and F. Shkurti, "Safe: Multitask failure detection for vision-language-action models," 2025.
- [20] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," in *European Control Conference*, 2019, pp. 3420–3431.
- [21] A. Taylor, A. Singletary, Y. Yue, and A. Ames, "Learning for safety-critical control with control barrier functions," in *Learning for Dynamics and Control*, 2020.
- [22] A. Agrawal and K. Sreenath, "Discrete control barrier functions for safety-critical control of discrete systems with application to bipedal robot navigation," in *Robotics: Science and Systems*, 2017.
- [23] P. Valle, C. Lu, S. Ali, and A. Arrieta, "Evaluating uncertainty and quality of visual language action-enabled robots," 2025.
- [24] J. Zhang, P. Fiers, K. A. Witte, R. W. Jackson, K. L. Poggensee, C. G. Atkeson, and S. H. Collins, "Human-in-the-loop optimization of exoskeleton assistance during walking," *Science*, vol. 356, no. 6344, pp. 1280–1284, 2017.
- [25] K. L. Poggensee and S. H. Collins, "How adaptation, training, and customization contribute to benefits from exoskeleton assistance," *Science Robotics*, vol. 6, no. 58, p. eabf1078, 2021.
- [26] A. Christou, A. Sochopoulos, E. Lister, and S. Vijayakumar, "Human-in-the-loop optimisation in robot-assisted gait training," 2025.
- [27] International Organization for Standardization, "Iso 13482:2014: Robots and robotic devices—safety requirements for personal care robots," 2014, current edition at the time of writing; revision in progress.
- [28] —, "Iso 14971:2019: Medical devices—application of risk management to medical devices," 2019.
- [29] U.S. Food and Drug Administration, "Product classification: Powered lower extremity exoskeleton, product code phl, 21 cfr 890.3480," 2026, accessed June 2026.